

4. Nuclear waste materials give out alpha, beta and gamma radiations and are classified according to their level of activity. They fall into three groups as follows:

Group	Information
Low level waste	Contain relatively low levels of activity and consist of alpha emitters and low activity beta and gamma emitters.
Intermediate level waste	Have higher levels of activity than low level waste but are not as dangerous as high level waste.
High level waste	Produce a significant temperature rise as a result of their high activity. This has to be considered when planning waste storage or disposal methods.

- (a) Outline the dangers to people's health from each of alpha, beta and gamma radiations from sources that are external to the body. [3]

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- (b) Low level waste can, after some basic treatment, be dumped in approved local authority sites. However, intermediate and high level wastes pose a much larger threat to life. They must be stored or disposed of in a way that keeps them well away from life forms for a very long time, typically tens of thousands of years or more.

The table below gives some data about four countries that generate significant amounts of high level waste from nuclear power stations.

Country	U.K.	Sweden	U.S.A.	France
Land area (km ²)	241 930	407 310	9 147 420	547 557
Population	65 637 240	9 903 120	323 127 510	66 689 000
Mean population per km ²	271.7	24.3	35.3	121.8
Number of nuclear reactors	15	8	99	58
Volume of low level waste in storage (m ³)	30 100	No information available	52 800	87 000
Volume of intermediate level waste in storage (m ³)	99 000		416 060	40 000
Volume of high level waste in storage (m ³)	1960			2 700

- (i) A person has suggested that for the U.K. and France, the volume of low level waste increases in proportion to the number of nuclear reactors in the country. Explain whether the suggestion is valid. [2]
Space for calculations.

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- (ii) Sweden, U.S.A. and France are well advanced in their planning for the long term storage of intermediate and high level wastes in deep underground storage facilities.

In 2010, the scientific director of the French authority for the long term storage of nuclear waste visited the U.S.A. storage site at Yucca Valley. Following his visit, he made the statement "There are far better sites in the U.S.A. than at Yucca mountain. When I went to the top of Yucca mountain and saw the volcanoes below, that worried me".

The development of the Yucca site has been stopped since around that time despite the spending of \$10 billion on two decades of work. The Americans are still investigating an alternative site to Yucca Valley.

- I. Explain the reason for concern over the choice of the Yucca Valley for the long term storage of nuclear waste. [2]

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- II. Use information in the table opposite to suggest a reason why a country such as the U.K. finds it difficult to agree on a site for a deep storage facility. [2]

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- (iii) The U.K. opened the world's first nuclear power station in 1956 in Calder Hall in Windscale amidst claims that, in the future, electricity would be free. The present view is that, with everything taken into account, electricity production from nuclear fuel is unacceptably expensive. Explain why there has been a change of view. [2]

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- (iv) Discuss an advantage and a disadvantage of **one** option, other than deep underground storage, for the long term disposal of dangerous nuclear waste. [2]

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